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(54) **SURGERY PLATFORM WITH A ROTATING SURFACE ON A STATIONARY BASE WITH INTEGRATED GAS LINES, INTEGRATED HEATING ELEMENTS, AND LEG POSITIONING STRAP SYSTEM**

(71) Applicant: **University of Kentucky Research Foundation**, Lexington, KY (US)

(72) Inventors: **Noe Tirado-Muniz**, Lexington, KY (US); **Mark A. Suckow**, Lexington, KY (US)

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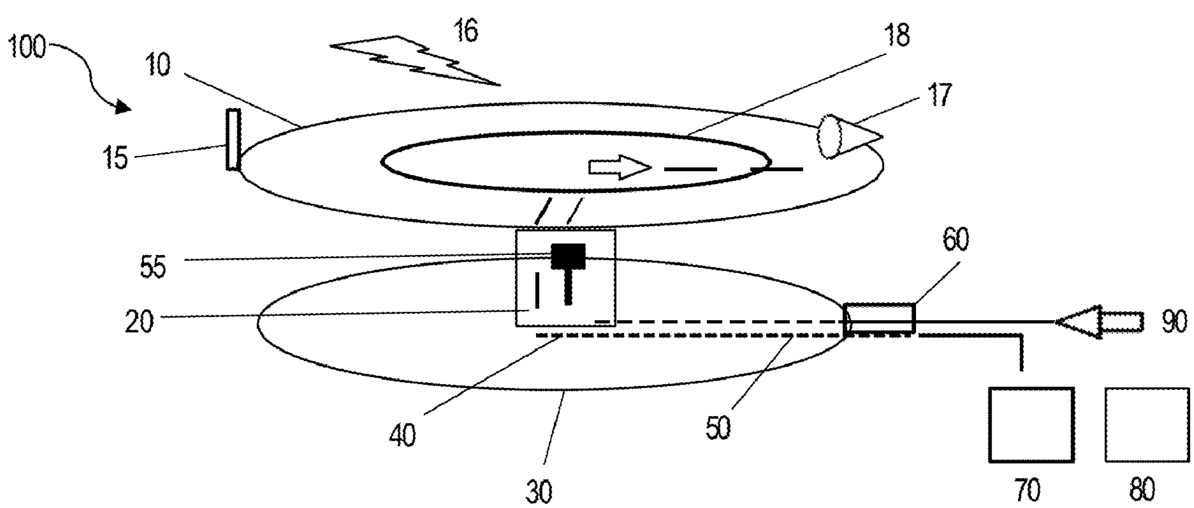
(57) **ABSTRACT**

The present disclosure concerns a surgery platform with a rotating surgery surface on a stationary base with integrated transport lines for feeding power and gas to the surgical surface. In some aspects, the platform can be utilized for surgery on small animals, such as rodents. The system can include a lower stationary base which supports the rotating upper operating surface via a pillar. The integrated lines run through the base into the center pillar and up to a swivel mechanism that allows rotation without kinking/twisting of either.



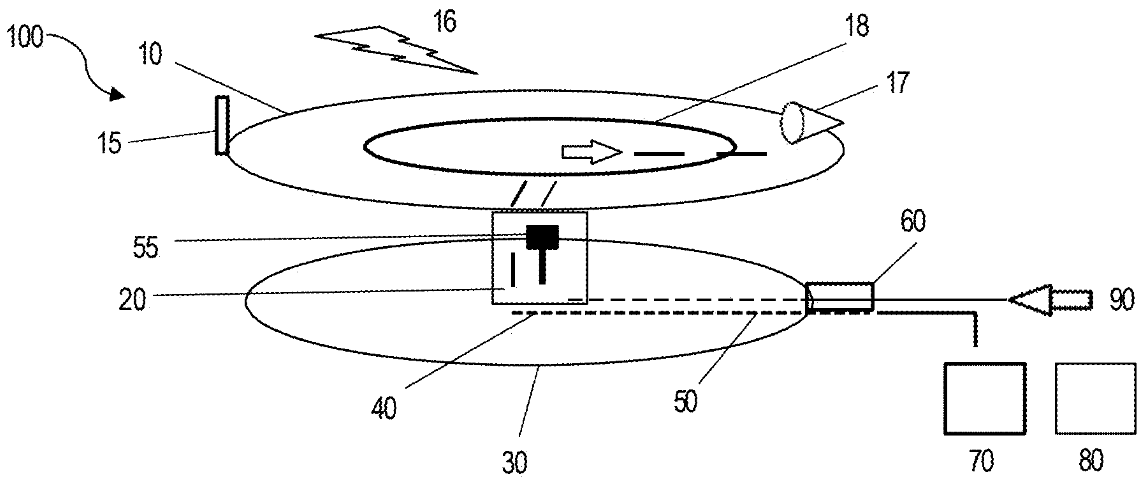


FIG. 1

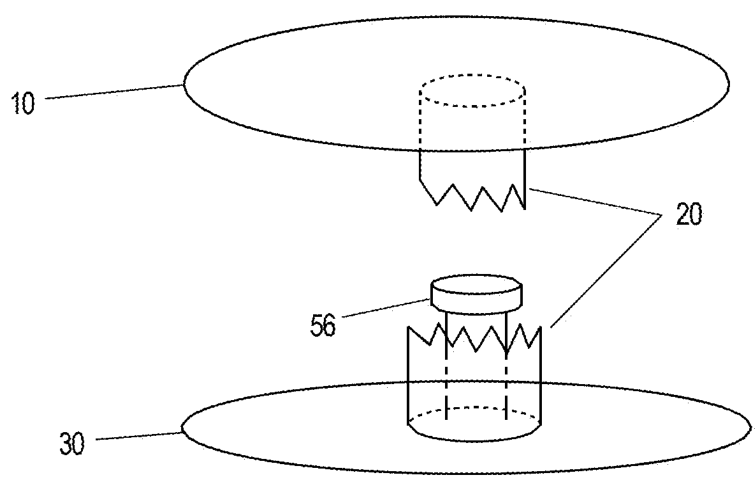


FIG. 2

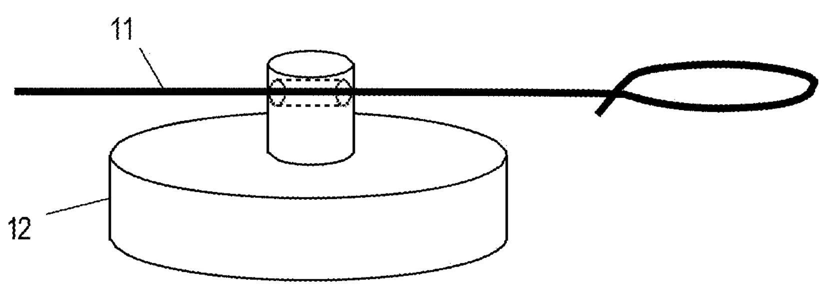


FIG. 3

SURGERY PLATFORM WITH A ROTATING SURFACE ON A STATIONARY BASE WITH INTEGRATED GAS LINES, INTEGRATED HEATING ELEMENTS, AND LEG POSITIONING STRAP SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Application 63/557,188, filed Feb. 23, 2024, the content of which is hereby incorporated by reference in its entirety.

FIELD

[0002] This present disclosure relates to systems and methods for retaining small animals during surgical procedures.

BACKGROUND

[0003] Surgeries require access to limited space within a subject. The type of surgery can require the angle of the surgeon to change as the procedure proceeds. Further, the access and subject size can severely hinder the surgeon's ability to adequately access the desired sites. Peripheral items, such as lighting, heating, and anesthesia to the patient are often necessary, but can crowd the surgeon's space. Thus there is a need for an adaptable surgical system that allows for improved access.

SUMMARY

[0004] A 1st aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns a system for the surgery of small vertebrates comprising a stationary base, a rotatable upper surgery platform, and a pillar, wherein the pillar extends upward from the base and connects to a lower surface of the rotatable upper surgery platform.

[0005] A 2nd aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 1st aspect, further comprising a transport line, wherein the transport line comprises at least one of a gas line and/or an electric line and wherein the transport line passes from a distal end through the base and the pillar to a proximal end at the rotatable upper surgery platform.

[0006] A 3rd aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 2nd aspect, further comprising a swivel deposited along the transport line.

[0007] A 4th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 3rd aspect, wherein the swivel is located within the body of the pillar.

[0008] A 5th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 2nd aspect, wherein the transport line is a gas line and is operably connected to an anesthetic gas and/or oxygen gas source at the distal end and a nose cone and/or an endotracheal tube at the proximal end.

[0009] A 6th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 2nd aspect, wherein the transport line is an electrical line and is operably connected at the distal end to an electric source of current.

[0010] A 7th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 6th aspect, wherein the electrical line operably connected at the proximal end to a circulating water pump, a heating gel or an infrared heating pad on the upper surface of the rotatable upper surgery platform.

[0011] An 8th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 2nd aspect, further comprising a water line.

[0012] A 9th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 1st aspect, wherein the rotatable upper surgery platform is configured to rotate in a circular direction about a circumference of the pillar.

[0013] A 10th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 9th aspect, wherein the rotatable upper surgery platform is operably rotated by an electronic or pneumatic foot pedal.

[0014] An 11th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 9th aspect, wherein the rotatable upper surgery platform is operably tiltable in a vertical or horizontal direction.

[0015] A 12th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 1st aspect, wherein the rotatable upper surgery platform further comprises an adjustable strap operably connected to a magnet on the upper surface.

[0016] A 13th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 1st aspect, further comprising at least one of one or more mirrors positioned around a perimeter of the upper surface of the rotatable upper surgery platform, a light source, a magnification source, a pulse oximeter, an electrocardiogram, or a combination thereof.

[0017] A 14th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 1st aspect, wherein a braking mechanism is operably connected to the rotatable upper surgery platform to prevent rotation and or tilting or to maintain a position of the rotatable upper surgery platform.

[0018] A 15th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 1st aspect, further comprising one or more image capturing devices positioned on the upper surface of the rotatable upper surgery platform.

[0019] A 16th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the system of the 1st aspect, wherein the pillar is extendible to thereby raise or lower the rotatable upper surgery platform.

[0020] A 17th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns a method for performing comprising placing a subject on the system of claim **1** and rotating the rotatable upper surgery platform about the pillar.

[0021] An 18th aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the method of the 17th aspect, wherein the system further comprises a gas line and an electric line with a swivel disposed along each, wherein the gas line and is operably connected to an anesthetic gas and/or oxygen gas source and wherein the electric line is operably connect to a power source.

[0022] A 19^{th} aspect of the present disclosure, either alone or in combination with any other aspect herein, concerns the method of the 18^{th} aspect, wherein the electric line is operably connected to a heating pad on the rotatable upper surgery platform.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. **1** shows a rotating surgery surface on a stationary base with an integrated anesthetic/O_2 gas line, integrated electrical heating element, magnetic limb positioning strap system and optional light source and/or magnification.

[0024] FIG. **2** shows the system including a lower stationary base which supports the rotating upper operating surface via a pillar.

[0025] FIG. **3** shows the adjustable strap and anchoring that allows for repositioning/adjustment for the subject size and/or needs.

DESCRIPTION

[0026] The present disclosure concerns systems and methods to provide a rotatable surgical platform for small animals. In some aspects, the present disclosure concerns a surgery platform with a rotating surgery surface on a stationary base with integrated anesthetic/O_2 gas lines, integrated electrical heating elements, magnetic leg positioning strap system and optional light source and magnification (FIG. **1**). The present disclosure concerns a surgery platform with a rotating surgery surface on a stationary base with integrated anesthetic/O2 gas lines, integrated electrical heating elements, magnetic leg positioning strap system and optional light source and magnification (see, e.g., FIG. **1**). In some aspects, the platform can be utilized for surgery on rodents. The system can include a lower stationary base which supports the rotating upper operating surface via a pillar (FIG. **2**). The base includes a port to attach both anesthetic gas/oxygen and electrical power supply. The integrated lines (e.g. anesthetic gas and power) run through the base into the center pillar and up to a swivel mechanism that allows rotation without kinking/twisting of either the gas or the electrical components. The swivel system connects integrated anesthetic gas and electrical power lines to the upper rotating operating platform. The power line feeds either a water (via a circulating water pump), gel "bed" or infrared pad located towards the center of the platform to keep the animal warm during the procedure. The gas line delivers anesthetic and oxygen to the patient via a nose cone or endotracheal mask/tube. The operating platform has a detachable handle made of autoclavable material and which can be used to rotate the platform. The platform may also be rotated by means of an electronic foot-pedal. In addition to rotating in a circular plane, the platform may also be tilted vertically or horizontally. Any rotation of the platform includes a hold mechanism so that the orientation of the platform is held in place while being used. The patient is placed on the rotating table and held in position using straps made of materials such as nylon and anchored by magnets that can be repositioned/adjusted depending on the patient size and procedure needs (FIG. **3**). This system would allow the surgeon to rotate the patient in a circular plane for ease of manipulation while maintaining the sterile field, without disrupting the constant delivery of anesthetic gas/O2. It also allows the surgeon an easier approach to rodent surgery in a more ergonomic fashion decreasing fatigue, improving productivity and potentially decreasing negative surgery outcomes.

[0027] Turning to FIG. **1**, the system **100** includes a rotatable upper surgery platform **10** positioned on top of a pillar **20** that in turn extends from a stationary base **30**. The pillar **20** connects to a lower surface of the rotatable upper surgery platform **10** in a manner such that the rotatable upper surgery platform **10** rotates about a central y axis through the pillar **20**. In some aspects, the rotatable upper surgery platform **10** is of a circular shape, though such is not required, but merely renders rotation balanced. The stationary base **30** which supports the rotatable upper surgery platform **10** via the pillar **20**. The stationary base **30** includes a port **60** to attach both anesthetic gas/oxygen and electrical power supply. In some aspects, the rotatable upper surgery platform **10** can be raised and/or lowered, such as through the pillar **20** being configured to extended in length.

[0028] In some aspects, the system and methods of the present disclosure provide for transport lines to run from the exterior of the system to the rotatable upper surgery platform **10**. Transport lines may include a gas line **50** and/or an electric line **40** or similar, such as a water line (not shown). Connected to the stationary base **30** through a connection port **60** are a gas line **50** and an electric line **40**, as well as any other transport lines, that feed through the stationary base **30** and up through the interior of the pillar **20** to reach the rotatable upper surgery platform **10**. In order to avoid kinking in the gas line **50** and/or the electric line **40** as the rotatable upper surgery platform **10** rotates about the pillar **20**, one or more swivel(s) **55** can be introduced along the length of each line to thereby effectively allow the gas line **50** and the electric line **40** to freely rotate about the same y-axis as the rotatable upper surgery platform **10**.

[0029] The swivel(s) **55** in the gas line **50** and/or the electric line **40** allow for the flow of gas and/or electricity to an upper surface of the rotatable upper surgery platform **10** from stationary sources outside or beside the system **100**. It is a difficulty in translating a need for gas and/or electrical power to a small animal surgical surface while still being able to provide rotation thereof, the options are to either provide the gas and/or power on the surgical surface or to severely limit the rotation of the surgical surface. The placement of a swivel **55** allows both obstacles to be overcome, freeing the surgery surface for access to the subject on the upper surface thereof. The integrated lines (e.g. anesthetic gas and power) run through the base into the center pillar and up to a swivel mechanism that allows rotation without kinking/twisting of either the gas or the electrical components.

[0030] In some aspects, the gas line **50** can run a gas source **80** through a control **90** and connection port **60** into the system **100**, wherein it runs through the stationary base **30** through the interior of the pillar **20** and up to the rotatable upper surgery platform **10** through a swivel **55**. On the upper surface of the rotatable upper surgery platform **10**, the gas line **50** can be operably connected to a breathing apparatus **17** such as an intubation tube or a breathing mask to supply the gas from the gas line **50** to the subject on the rotatable upper surgery platform **10**. The gas of the gas line **50** can be oxygen (O_2), an anesthetic gas, or a combination. The gas can include air or filtered air. The gas can be pressurized.

The gas line **50** can optional include common features such as pressure regulators, valves, and such further along the length thereof.

[0031] Similarly, the electric line **40** connects a power source **70** to the rotatable upper surgery platform **10** in a similar pathway. The electrical line **40** may be operably connected to one or more end points on the rotatable upper surgery platform **10**, such as an integrated heating element **18**. In some aspects, the electric line **40** can be operably connected to a water pump. Additional transport lines such as a water line (not shown) can similarly use a swivel **55** to provide water to the rotatable upper surgery platform **10**. A water line can be operably connected to a pump to provide a stream of water. A water line can be operably connected to a valve to control the flow of water.

[0032] Turning to FIG. **2**, shown a cut away view of the pillar **20** with a swivel platform **56** in the interior space therein. The swivel platform **56** provides a plate upon which the rotatable upper surgery platform **10** can rotate. Thus the pillar **20** need not rotate itself, but instead provide the structural support to the rotatable upper surgery platform **10** while the swivel platform **56** provides the axis of rotation.

[0033] Turning back to FIG. **1**, the rotatable upper surgery platform **10** can be of a sufficiently small surface area as needed for the subject being operated on. The transport lines of the gas line **50** and the electric line **40**, being able to be connected to off platform sources (via the swivel(s) **55**) frees the surface area. The

[0034] A system for the surgery of small vertebrates comprising a stationary base, a rotatable upper surgery platform, and a pillar, wherein the pillar extends upward from the base and connects to a lower surface of the rotatable upper surgery platform. Rotation of the rotatable upper surgery platform **10** can be either manually or electrically or pneumatically controlled, or through a combination thereof. For example, a handle **15** can be placed on the rotatable upper surgery platform **10** to allow for the rotation thereof about the pillar **20** and/or swivel platform **56** therein. In some aspects, the handle **15** is detachable such that it can be sterilized and/or cleaned, such as with an autoclave. A device such as a foot pedal (not shown) can connect pneumatic or electric energy to a motor (not shown) operably connected to the rotatable upper surgery platform **10** (such as through the swivel platform **56**) to cause rotation thereof. It is a further feature that the rotatable upper surgery platform **10** be operably connected to a braking or rotation damping system (not shown) to cause the rotation thereof to halt and/or to maintain the rotatable upper surgery platform **10** in a particular position.

[0035] In some aspects, the rotatable upper surgery platform **10** can be adjusted additionally to tilt either vertically and/or horizontally. In further aspects, the system may include means to maintain the rotatable upper surgery platform **10** in a desired titled position.

[0036] In some aspects, the rotatable upper surgery platform **10** can feature additional features on or about the upper surface thereof. Such may include a camera or similar image capture device (not shown). Such may include a light source (not shown). Such may include a source for magnification (not shown) to allow a user to better visualize the surgical procedure(s) on the subject. Such may include one or more surgical instrument holders and/or rests (not shown). Such may include connections or the ability to receive additional surgical monitoring systems (not shown) such as a pulse oximeter, an ECG (electrocardiogram) machine or leads thereto.

[0037] In some aspects, the rotatable upper surgery platform **10** may be made at least in part of a ferromagnetic material, such as a steel, including surgical steel. As shown in FIG. **1**, such can offer magnetic attachments **16** to the upper surface of the rotatable upper surgery platform **10**. In some aspects, a magnet can be used to assist in restraining the subject during a surgical procedure.

[0038] Turning to FIG. **3**, shown is a magnetic restraint to limit the movement of a subject's limb during a procedure. The restrain includes a magnet **12** and a loopable material **11** that affixes to the magnet **12**. For example, as depicted, the magnet **12** includes a hole through which the loopable material **11** can be threaded. The loopable material **11** can be looped around the limb of a subject (not shown) to effectively tie the limb to the magnet **12**. The attracting between the magnet **12** and the ferromagnetic material of the rotatable upper surgery platform **10** allows for the limb to effectively be tied to the rotatable upper surgery platform **10**. It will be appreciated that use of multiple such restraints will allow for multiple limbs to be retrained. For example, four of such restraints will allow for all four limbs of a small mammal to be restrained.

[0039] The present disclosure also includes methods of using the system to perform a surgical procedure on a subject. While the subject can be any animal, it is a distinct advantage that through the integrated transport lines of the present system, the system can accommodate small animals, such as small mammals, including rodents and the like. The present system allows the surgeon to rotate the patient in a circular plane for ease of manipulation while maintaining the sterile field, without disrupting the constant delivery of anesthetic gas/O_2. The present system also allows the surgeon an easier approach to surgery on small animals in a more ergonomic fashion decreasing fatigue, improving productivity and potentially decreasing negative surgery outcomes.

[0040] In some aspects, the methods include use of the magnet(s) **12**, the loopable material **11**, and the rotatable upper surgery platform **10**. A limb of the subject is looped through the loopable material **11** and the magnet **12** adheres to the upper surface of the rotatable upper surgery platform **10**.

[0041] In some aspects, the methods include flowing a gas such as an anesthetic gas and/or O_2 through the gas line **50**. At a proximal end of the gas line **50** on the rotatable upper surgery platform **10**, the gas line **50** is operably connected to a breathing apparatus such as an intubation tube or a mask that delivers the gas to the subject.

[0042] In some aspects, the methods include providing heat to the subject during a surgical procedure. For example, a proximal end of the electrical line **40** is operably connected to a heating pad **18** on the rotatable upper surgery platform **10**. The heating pad **18** thereby provides heat to the body of a subject on top thereof.

[0043] Various modifications of the present disclosure, in addition to those shown and described herein, will be apparent to those skilled in the art of the above description. Such modifications are also intended to fall within the scope of the appended claims.

[0044] It is appreciated that all reagents are obtainable by sources known in the art unless otherwise specified.

[0045] It is also to be understood that this disclosure is not limited to the specific aspects and methods described herein, as specific components and/or conditions may, of course, vary. Furthermore, the terminology used herein is used only for the purpose of describing particular aspects of the present disclosure and is not intended to be limiting in any way. It will be also understood that, although the terms "first," "second," "third" etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, "a first element," "component," "region," "layer," or "section" discussed below could be termed a second (or other) element, component, region, layer, or section without departing from the teachings herein. Similarly, as used herein, the singular forms "a," "an," and "the" are intended to include the plural forms, including "at least one," unless the content clearly indicates otherwise. "Or" means "and/or." As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms "comprises" and/or "comprising," or "includes" and/or "including" when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof. The term "or a combination thereof" means a combination including at least one of the foregoing elements.

[0046] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0047] Reference is made in detail to exemplary compositions, aspects and methods of the present disclosure, which constitute the best modes of practicing the disclosure presently known to the inventors. The drawings are not necessarily to scale. However, it is to be understood that the disclosed aspects are merely exemplary of the disclosure that may be embodied in various and alternative forms. Therefore, specific details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for any aspect of the disclosure and/or as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

[0048] Patents, publications, and applications mentioned in the specification are indicative of the levels of those skilled in the art to which the disclosure pertains. These patents, publications, and applications are incorporated herein by reference to the same extent as if each individual patent, publication, or application was specifically and individually incorporated herein by reference.

[0049] The foregoing description is illustrative of particular embodiments of the disclosure, but is not meant to be a limitation upon the practice thereof. The following claims, including all equivalents thereof, are intended to define the scope of the disclosure.

We claim:

1. A system for the surgery of small vertebrates comprising a stationary base, a rotatable upper surgery platform, and a pillar, wherein the pillar extends upward from the base and connects to a lower surface of the rotatable upper surgery platform.

2. The system of claim **1**, further comprising a transport line, wherein the transport line comprises at least one of a gas line and/or an electric line and wherein the transport line passes from a distal end through the base and the pillar to a proximal end at the rotatable upper surgery platform.

3. The system of claim **2**, further comprising a swivel deposited along the transport line.

4. The system of claim **3**, wherein the swivel is located within the body of the pillar.

5. The system of claim **2**, wherein the transport line is a gas line and is operably connected to an anesthetic gas and/or oxygen gas source at the distal end and a nose cone and/or an endotracheal tube at the proximal end.

6. The system of claim **2**, wherein the transport line is an electrical line and is operably connected at the distal end to an electric source of current.

7. The system of claim **6**, wherein the electrical line operably connected at the proximal end to a circulating water pump, a heating gel or an infrared heating pad on the upper surface of the rotatable upper surgery platform.

8. The system of claim **2**, further comprising a water line.

9. The system of claim **1**, wherein the rotatable upper surgery platform is configured to rotate in a circular direction about a circumference of the pillar.

10. The system of claim **9**, wherein the rotatable upper surgery platform is operably rotated by an electronic or pneumatic foot pedal.

11. The system of claim **9**, wherein the rotatable upper surgery platform is operably tiltable in a vertical or horizontal direction.

12. The system of claim **1**, wherein the rotatable upper surgery platform further comprises an adjustable strap operably connected to a magnet on the upper surface.

13. The system of claim **1**, further comprising at least one of one or more mirrors positioned around a perimeter of the upper surface of the rotatable upper surgery platform, a light source, a magnification source, a pulse oximeter, an electrocardiogram, or a combination thereof.

14. The system of claim **1**, wherein a braking mechanism is operably connected to the rotatable upper surgery platform to prevent rotation and or tilting or to maintain a position of the rotatable upper surgery platform.

15. The system of claim **1**, further comprising one or more image capturing devices positioned on the upper surface of the rotatable upper surgery platform.

16. The system of claim **1**, wherein the pillar is extendible to thereby raise or lower the rotatable upper surgery platform.

17. A method for performing comprising placing a subject on the system of claim **1** and rotating the rotatable upper surgery platform about the pillar.

18. The method of claim **17**, wherein the system further comprises a gas line and an electric line with a swivel disposed along each, wherein the gas line and is operably connected to an anesthetic gas and/or oxygen gas source and wherein the electric line is operably connect to a power source.

19. The method of claim **18**, wherein the electric line is operably connected to a heating pad on the rotatable upper surgery platform.

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